VigilAI BorderShield - Production System Blueprint

AI video analytics for existing border CCTV infrastructure | Exact SIH scope mapped to an edge-first product

Existing CCTV estate
SOURCE

- RTSP / ONVIF Profile T
- IP cameras, NVR and fixed posts
- No mandatory smart-camera replacement

streams

Remote edge appliance
EDGE

- Decode, buffer and health checks
- GPU inference and local rules
- Store-and-forward during link loss

frames

Vision intelligence
AI

- Person and vehicle tracking
- ANPR, face detection, night movement
- Virtual fence and suspicious activity

events

Trusted event fabric
EVENTS

- Signed events and evidence hashes
- Priority, dedupe and retries
- Policy-controlled retention

alerts

Command
HMI

- Live map
- Alert triage
- Audit

What the product solves

- Converts standard CCTV feeds into actionable intelligence without dedicated FRS, ANPR or smart-camera hardware.
- Detects and tracks people and vehicles, monitors virtual fences, analyzes night movement and raises real-time incidents.
- Operates at remote posts with weak connectivity and synchronizes signed event evidence when the link returns.

Professional accuracy position

- No responsible computer-vision product can guarantee 100% accuracy on unseen real-world footage.
- The system reports measured precision, recall, F1, mAP, IDF1, false-alert rate and latency for each camera condition.
- 100% commitments apply to audit traceability, schema validation, test evidence and recovery of acknowledged events.

Judge-ready product statement

VigilAI BorderShield is an offline-first, software-defined border video intelligence platform that upgrades existing CCTV into a measurable, auditable and scalable surveillance network.

Exact SIH26-26187 Problem and Product Response

Organization: Ministry of Home Affairs | Category: Software | Theme: Smart Automation

Problem in the field

- Border Out Posts, check posts, roads and strategic sites already use conventional CCTV.
- Continuous human observation is required because most cameras only record and display video.
- FRS, ANPR, tracking and intrusion functions often need costly proprietary hardware.
- Remote links, darkness, weather and large camera estates increase operating difficulty.

Capabilities explicitly required

- Human detection and tracking; vehicle detection and classification.
- Face detection; Automatic Number Plate Recognition.
- Virtual-fence intrusion and suspicious-activity detection.
- Night movement detection, real-time alerts and event logging.
- Integration with existing command and control systems.

Expected result

- Software-defined intelligence on standard IP camera streams.
- Lower dependence on expensive dedicated surveillance hardware.
- Faster situational awareness and incident response.
- Scalable, cost-effective deployment across remote locations.
- Actionable alerts with reviewable evidence and audit history.

SIH requirement	VigilAI production response	Evidence for judges
Existing CCTV	RTSP/ONVIF ingest adapters and camera discovery	Live multi-stream demo + camera health
Human / vehicle	Detector + tracker + cross-camera event identity	Precision/recall and IDF1 report
ANPR / face	Authorized analytics microservices with confidence gates	Day/night test matrix; privacy controls
Virtual fence	Geofenced rules, persistence and direction logic	Replay intrusion with alert latency
Suspicious activity	Hybrid temporal model + deterministic rules	Defined behavior classes and F1
Night movement	Low-light pipeline, camera quality gating	Night dataset recall and false alerts
Real-time alerts	Priority queue, dedupe, evidence, acknowledgements	End-to-end P95 latency dashboard
Remote deployment	Edge-first inference and durable store-forward	Link-loss and recovery demonstration

Current VigilAI Repository - Reuse Map

Audited commit 0690c45 | Strong PoC patterns exist, but border production services are not yet implemented

25-35%
ESTIMATED DEMO FOUNDATION REUSE
Architecture estimate, not measured completion

10-20%
ESTIMATED PRODUCTION SCOPE
Border-specific and operational hardening remain

0
REAL BORDER FIELD VALIDATIONS
No representative operational dataset yet

6
CORE REUSABLE PATTERNS
Detection, tracking, camera, rules, evidence, UI

Reuse with refactoring

- YOLO detection and ByteTrack tracking patterns from Vigil Exam / Vigil Traffic.
- Camera registration, per-camera API key and heartbeat concept.
- Rule and decision engine patterns with persistence and cooldown.
- Evidence capture, alert APIs, PostgreSQL models and React dashboards.
- Vehicle counting and early speed-estimation experiments.

Must be replaced or hardened

- Single-file camera loops and hardcoded localhost configuration.
- Local webcam/video input instead of managed RTSP/ONVIF streams.
- Simple pixel-to-speed conversion without camera calibration.
- Unversioned AI artifacts and no reproducible benchmark pipeline.
- No containers, CI gates, observability, SBOM or signed releases.

New border-grade modules

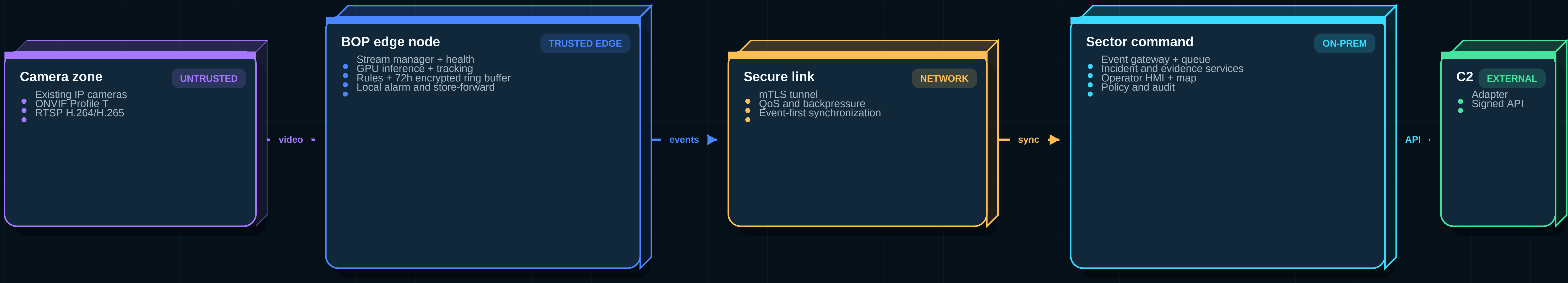
- Edge stream gateway, local encrypted buffer and store-forward sync.
- ANPR, face detection/authorized matching, night and low-light analytics.
- Virtual fence, loitering, direction and suspicious-activity services.
- Incident workflow, geospatial command map and command-center adapter.
- MLOps, security hardening, model drift and field validation.

Do not present the current repository as a finished border product

Current artifact	Production interpretation	Required upgrade
Vigil Exam classroom PoC	Useful multi-person detection / tracking pattern	Decouple into edge services; remove exam-specific logic
Vigil Traffic scripts	Vehicle detector/tracker starter	Calibration, tests, stream manager, model registry
Express + Prisma APIs	Reusable CRUD and alert ideas	Typed contracts, OIDC/RBAC, audit, event-driven design
React dashboards	Reusable UX components	Map, incident workflow, WebSocket, accessibility, offline state
API-key camera auth	Good prototype concept	Device certificates, rotation, mTLS and secure provisioning

Production Edge-to-Command Architecture

Local decisions continue during network loss; central systems receive compact, signed events instead of continuous raw video



Edge invariants

- Inference, virtual-fence decisions and local alerts do not depend on cloud availability.
- Raw video remains local by default; only incident clips and metadata synchronize.
- Every accepted event has eventId, cameraId, modelVersion, timestamps and SHA-256 evidence hash.

Link-loss behavior

- Durable event outbox, bounded local storage and priority-based synchronization.
- Critical incidents sync before routine health telemetry after connectivity returns.
- Duplicate delivery is safe through idempotency keys and acknowledgement checkpoints.

Central control

- Sector-wide correlation, geospatial map, incident assignment and review workflow.
- Configuration and model releases are signed, staged and remotely rolled back.
- Command integration uses a versioned adapter; core services remain vendor-neutral.

Deployment mode	Use	Compute profile	Connectivity
Pilot	1-4 cameras at one post	Single x86/NVIDIA or Jetson-class edge node	LAN + optional WAN
BOP production	4-16 streams per post	Redundant edge pair; GPU capacity benchmarked	Intermittent secure link
Sector command	Hundreds of event sources	HA container cluster + object storage	Private network
Central C2	Cross-sector intelligence	Adapter only; no raw-video dependency	Policy-controlled API

Per-Camera AI Pipeline - Deterministic First, ML Where Needed

A staged pipeline lowers GPU cost and false alerts while preserving evidence and model traceability

Ingest 1
- RTSP reconnect
- Clock sync
- Decode

Quality 2
- Blur / dark
- Tamper
- FPS health

Detect 3
- Person
- Vehicle
- Plate / face

Track 4
- ByteTrack
- Trajectory
- Re-ID gate

Context 5
- Zones
- Direction
- Persistence

Behavior 6
- Intrusion
- Loitering
- Night motion

Decision 7
- Confidence
- Dedupe
- Priority

Event OUT
- Clip
- Hash
- Alert

Fast path - every frame

- Stream continuity, quality checks, decode telemetry and lightweight motion gating.
- Detector and tracker run at a profiled frame rate; display can remain at source FPS.
- Camera condition becomes part of every model metric and alert decision.

Context path - every track

- Polygon zones, virtual fences, entry direction, dwell time and object association.
- Rules require temporal persistence rather than a single-frame detection.
- Cross-camera Re-ID is optional and used only when operationally justified.

Review path - every incident

- Alert includes pre-event and post-event clip, annotated keyframe and reason codes.
- High-risk face/watchlist results require a second signal or human confirmation.
- Operator feedback becomes labeled data, never an automatic ground truth.

Analytics	Primary method	False-alert control	Output
Human / vehicle	License-vetted detector exported to ONNX/TensorRT	Class thresholds + tracking persistence	Track events
ANPR	Plate detector + perspective correction + OCR	Regex, region formats, multi-frame voting	Plate candidate
Face	Face detector; authorized verifier only	Quality gate + FAR threshold + human review	Match candidate
Virtual fence	Track-line geometry and direction	Minimum dwell and crossing persistence	Intrusion event
Suspicious activity	Rules + temporal classifier	Defined classes, context and confidence fusion	Behavior event
Night movement	Low-light quality gate + motion + detector	Scene-specific thresholds and thermal input if available	Night event

Recommended Production Technology Stack

Pinned, license-reviewed and benchmarked releases; avoid changing frameworks during the finale

Layer	Primary stack	Why this choice	Production control
Camera interoperability	ONVIF Profile T + RTSP; GStreamer	Vendor-neutral H.264/H.265 ingest and resilient pipelines	Camera conformance matrix
Edge video runtime	NVIDIA DeepStream / GStreamer C++ plugins	Multi-stream decode, batching, tracking and GPU efficiency	Profile per hardware SKU
Inference serving	TensorRT + Triton; ONNX Runtime CPU fallback	Versioned multi-model serving, health and metrics	Signed model repository
Vision models	License-vetted RT-DETR/YOLO family + ByteTrack	Strong real-time baseline; exportable deployment format	Benchmark decides final model
ANPR	Plate detector + PaddleOCR/approved OCR	Perspective correction and multi-frame voting	Regional plate grammar
AI services	Python 3.12, FastAPI, gRPC, Pydantic	Typed ML boundaries and rapid team development	Contract tests
Control plane	NestJS + TypeScript + OpenAPI	Structured modules, validation, WebSocket and RBAC	Strict TypeScript
Command HMI	React + TypeScript + MapLibre + WebSocket	Live geospatial incidents and replay	Playwright tests
Event fabric	NATS JetStream at pilot; Kafka/Redpanda at sector scale	Durable ordered events and replay	Schema registry
Operational data	PostgreSQL + TimescaleDB + PostGIS	Transactions, telemetry and spatial queries	HA backups / PITR
Evidence storage	MinIO S3-compatible object store	On-prem clips, lifecycle policy and object locking	SHA-256 manifest
Identity / secrets	Keycloak OIDC, mTLS, Vault-compatible secrets	MFA, RBAC and certificate rotation	No long-lived API keys
Runtime	Docker; K3s at edge cluster; Kubernetes at command	Reproducible rollout and rollback	Signed images + SBOM
Observability	OpenTelemetry, Prometheus, Grafana, Loki, Tempo	Metrics, logs and traces without vendor lock-in	SLO alerting
MLOps	CVAT, DVC, MLflow, data/model cards	Lineage, approvals, versioning and rollback	Champion/challenger
CI / security	GitHub Actions, Trivy, Semgrep, pytest, k6	Automated quality and supply-chain gates	Protected releases

Deployable Service Structure

Keep the pilot compact; split services only where scaling, security or failure isolation requires it

REMOTE EDGE SERVICES

edge-agent
- Camera discovery
- RTSP reconnect
- health + ring buffer

vision-runtime
- decode + inference
- tracking
- GPU scheduler

analytics-engine
- ANPR / face
- night
- behavior

event-engine
- zones + fences
- confidence fusion
- dedupe + priority

sync-agent
- durable outbox
- bandwidth policy
- ack + replay

local-hmi
- post status
- critical alarm
- offline review

signed events

SECTOR COMMAND SERVICES

camera-registry
- device identity
- config versions
- certificate lifecycle

incident-service
- workflow
- assignment
- status and SLA

evidence-service
- clips + keyframes
- hash manifest
- retention / redaction

alert-dispatcher
- WebSocket
- SMS/email adapters
- escalation policy

command-api
- OIDC + RBAC
- REST / WebSocket
- C2 adapter

command-ui
- map + timeline
- triage + replay
- audit views

Shared contracts
- CameraHealth, TrackEvent, IntrusionEvent, PlateCandidate, FaceCandidate and IncidentEvent.
- Protobuf for internal gRPC; JSON Schema/OpenAPI at external boundaries.
- Every event carries schemaVersion, eventId, timestamps, source and model lineage.

Failure isolation
- A failed analytics model cannot stop camera recording or stream health monitoring.
- Alert delivery retries do not block inference; durable outbox prevents silent loss.
- Command API failure does not prevent local BOP alarms or evidence buffering.

Pilot deployment rule
- Run edge services in 2-3 containers for the SIH demo, not twelve tiny containers.
- Preserve the same contracts so services can be separated at production scale.
- One repository, clear ownership, reproducible Compose and K3s manifests.

Event, Evidence and Storage Architecture

Event-first design minimizes bandwidth while maintaining complete traceability

Frame metadata
EPHEMERAL
- cameraId + frameId
- captureTime + ingestTime
- quality + calibration

TrackEvent
STREAM
- trackId + class
- bbox + trajectory
- confidence + model

IncidentEvent
DURABLE
- type + severity
- zone + reason codes
- correlationId

EvidenceManifest
IMMUTABLE
- clip URI + hash
- keyframe + annotations
- chain timestamps

AuditEvent
APPEND
- actor + action
- before / after
- requestId

Store	Data	Retention principle	Security
Edge ring buffer	Recent raw video per camera	24-72h configurable; overwrite oldest	Encrypted volume; local access only
PostgreSQL/PostGIS	Cameras, zones, incidents, operators	Policy based; relational source of truth	RBAC, backups, row-level controls
TimescaleDB	Health, latency, FPS, GPU and drift metrics	Downsample after operational window	No biometric media
MinIO	Incident clips, keyframes and manifests	Lifecycle policy; legal hold when authorized	Object lock, encryption, signed access
Event log	Track/incident/alert/audit events	Replay window per policy	mTLS, ACLs, schema compatibility
MLflow/DVC	Datasets, model lineage and approvals	Versioned for reproducibility	Restricted engineering access

Idempotency and ordering
- eventId is globally unique; consumers store the last acknowledged sequence.
- At-least-once delivery plus idempotent consumers prevents silent loss.
- Camera clock offset is measured; capture and ingest timestamps remain separate.

Evidence integrity
- SHA-256 of original clip, rendered keyframe and manifest.
- Annotation is stored separately so original evidence is never overwritten.
- Every export records operator, purpose, timestamp and checksum.

Privacy minimization
- Upload event clips instead of continuous raw video by default.
- Face matching only against an authorized watchlist and policy.
- Non-incident faces/plates can be redacted in operator exports.

Zero-Trust Security and Privacy Controls

Security is a product requirement, not a final-stage checklist

Camera trust zone — ZONE 1
- VLAN isolation
- no direct internet
- allowlisted edge

Edge trust zone — ZONE 2
- device certificate
- secure boot where available
- encrypted local media

Transport — ZONE 3
- mTLS 1.3
- certificate rotation
- replay protection

Command zone — ZONE 4
- OIDC + MFA
- RBAC/ABAC
- network policies

C2 — ZONE 5
- adapter
- least privilege

Control	Implementation	Acceptance evidence
Device identity	Per-edge certificate; secure provisioning; revocation	Unknown/revoked device rejected
Operator identity	OIDC, MFA, RBAC and session timeout	Role matrix + negative authorization tests
Transport	mTLS, private networking, certificate rotation	Packet inspection shows encrypted channels
Secrets	Vault-compatible store; no secrets in code/images	Secret scan and rotation drill
Evidence	Encryption, signed manifest, immutable original	Hash verification before/after export
Software supply chain	Pinned dependencies, SBOM, signed images/models	CI release attestation
Audit	Append-only actor/action/config/model logs	Every privileged action queryable
Biometrics	Authorized purpose, strict thresholds, human verification	Policy gate and access audit
Incident response	Runbooks, backups, recovery and revocation	Tabletop test + recovery timing

Deployment gate

Before any real field use: organization approval, privacy/security review, VAPT, camera-network assessment, retention policy and authorized biometric workflow must be completed. This blueprint does not claim certification.

Accuracy Plan - What Can Honestly Be Promised

All numbers below are target acceptance gates until measured on a representative, frozen validation dataset

>=98%
INTRUSION EVENT RECALL TARGET
Representative day/night/weather scenarios

>=95%
INTRUSION PRECISION TARGET
False alerts measured per camera-hour

<=2s
P95 CRITICAL ALERT LATENCY
Capture to operator notification at edge

100%
AUDIT TRACEABILITY TARGET
Accepted events include full lineage and hash

Capability	Primary metrics	Target gate	Test slices
Person / vehicle	Precision, recall, mAP50-95	>=95% P and >=95% R for operational classes	Day/night, near/far, rain/dust, occlusion
Tracking	IDF1, HOTA, ID switches	>=90% IDF1 target	Crowding, crossings, camera shake
Virtual fence	Event precision/recall, delay	>=95% precision; >=98% recall	Direction, dwell, partial occlusion
ANPR	Plate recall; exact-string accuracy	>=95% plate recall day; report night separately	Angle, blur, glare, script/format
Face detection	Recall by face size / pose	>=95% within defined operational envelope	Pose, illumination, occlusion
Face verification	TAR at fixed FAR	Target TAR >=95% at FAR <=0.1%	Authorized dataset; demographic slices
Night movement	Event recall; false alerts/hour	>=92% recall; scene-specific false-alert cap	IR, low light, insects, weather
Suspicious activity	Per-class F1; confusion matrix	>=90% F1 only for explicitly defined classes	Loitering, running, abandoned object
Reliability	Event loss, availability, recovery	0 acknowledged-event loss; >=99.9% connected uptime target	Link loss, restart, disk pressure

Judge wording - use this

- We do not claim universal 100% AI accuracy. We report scenario-wise measured metrics and uncertainty.
- High-risk alerts combine temporal persistence, multiple signals and operator verification.

Never say this

- The product is 100% accurate in every real border condition.
- Synthetic or demo-video accuracy proves field deployment readiness.

100% claims allowed after tests

- 100% schema validation for released contracts.
- 100% accepted-event audit lineage and deterministic hash verification.

Production ML Lifecycle and Model Governance

Model quality comes from representative data, leakage prevention, calibration, monitoring and controlled rollback

1 Capture

Authorized clips with camera, scene, weather, illumination and distance metadata.

2 Curate

Deduplicate, redact where required, balance rare events and record provenance.

3 Annotate

CVAT labels with dual review, adjudication and annotation-quality sampling.

4 Split

Group by camera/site/time so neighboring frames never leak across datasets.

5 Train

Reproducible configs, fixed seeds, data version and experiment lineage.

validation

6 Evaluate

Frozen scenario matrix, calibration, false-alert cost and latency benchmark.

7 Approve

Model card, security/license review and signed champion artifact.

8 Deploy

Canary by site/camera, shadow mode, rollback and operator feedback.

9 Monitor

Drift, quality, FPS, latency, alert rate and human-confirmation rate.

Mandatory dataset slices

- Day, dusk, night, IR, fog/rain/dust, glare, compression and dropped frames.
- Near/far subjects, occlusion, crowding, camera angles and scene motion.
- Known negatives: animals, foliage, shadows, insects, headlights and maintenance staff.

Release evidence

- Dataset card, model card, confusion matrix and per-slice metrics.
- Calibration curve, operating thresholds, latency/GPU benchmark and license manifest.
- Signed model checksum, rollback version and approved deployment scope.

Continuous monitoring

- Input-quality drift, embedding drift, class frequency and confidence distribution.
- Operator-confirmed vs rejected incidents by site and condition.
- Automatic rollback only for software health; model quality changes require review.

Offline-First Reliability, SRE and Capacity Engineering

A border product is judged by behavior during failure, not only by the happy-path demo

Failure	Required system behavior	Test / SLO
WAN link down	Continue inference, local alert and durable queue	24h replay test; 0 acknowledged-event loss
Camera stream down	Reconnect with backoff; mark health; notify operator	Detect within 15s target
GPU overload	Adaptive sampling; priority cameras; no memory crash	Sustained 2x burst and soak test
Model crash	Restart isolated worker; preserve stream/evidence pipeline	Recovery under 30s target
Disk pressure	Retention watermark; preserve critical incidents first	80/90/95% watermark tests
Duplicate delivery	Idempotent incident creation and evidence manifest	Replay same event 100 times
Clock drift	Record offset; retain capture and ingest timestamps	NTP loss simulation
Bad model release	Canary health gate and signed rollback	Rollback drill under 5 min target
Command API down	Local BOP operation continues; sync after recovery	Restart and replay test

99.9%
CONNECTED SERVICE AVAILABILITY
Target after redundant production deployment

0
ACKNOWLEDGED EVENT LOSS
Durable outbox + replay acceptance gate

<=15s
CAMERA-LOSS DETECTION
Target depends on heartbeat / stream policy

72h
EDGE VIDEO RING BUFFER
Configurable by storage and retention policy

Golden signals

- Stream FPS / reconnects; inference latency; event backlog; GPU/memory/disk; alert-delivery latency.

Observability

- OpenTelemetry traces, Prometheus metrics, structured logs, Grafana dashboards and runbook-linked alerts.

Capacity rule

- Claim camera capacity only after testing the exact codec, resolution, FPS, model set and hardware SKU.

Six-Member Professional Work Division

Each member owns one bounded module, contracts, tests, documentation and a judge explanation

1 Edge and Camera Platform — M1

ONVIF/RTSP ingest, camera registry, health, reconnect, ring buffer and store-forward. Stack: GStreamer, Python/C++, Redis/NATS, Docker.

2 Detection and Tracking — M2

Person/vehicle model benchmark, TensorRT export, tracking, trajectories and performance profiling. Stack: PyTorch, ONNX, TensorRT, DeepStream.

3 ANPR, Face and Night — M3

Plate pipeline, authorized face candidate, low-light quality and night movement. Stack: OpenCV, OCR, model serving, calibration.

4 Intrusion and Behavior — M4

Virtual fences, direction, dwell, suspicious-activity classes, confidence fusion and false-alert reduction. Stack: Python, geometry, temporal ML, rules.

5 Control, Data and Security — M5

Typed API, incidents, evidence, Postgres/PostGIS, identity, RBAC, audit and C2 adapter. Stack: NestJS, Prisma, Keycloak, MinIO.

6 HMI, Integration and SRE — M6

Command map, live alerts, replay, Compose/K3s, observability, CI, end-to-end tests and final demo. Stack: React, TS, WebSocket, OTel, Playwright.

Shared rule	Definition
Contract first	No module integration begins without versioned sample payloads and contract tests.
Definition of done	Code + unit tests + integration fixture + metrics + README + 3-minute explanation.
Daily integration	Merge small vertical slices; run one camera-to-alert path at least twice daily.
AI agent use	Allowed for code generation/review, but member must understand contracts, tests, metrics and failure modes.

From Current PoC to Production Candidate

The 36-hour finale demonstrates a hardened vertical slice; field readiness requires staged validation after the hackathon

Phase 0 - Contract freeze
BEFORE FINALE
- Event schemas
- Camera/zone config
- Golden clips
- Repo/CI skeleton

Phase 1 - SIH vertical slice
36 HOURS
- 2-4 live streams
- Person/vehicle + fence
- Event evidence
- Command HMI

Phase 2 - Product beta
4-8 WEEKS
- ANPR/night
- offline sync
- security baseline
- benchmark suite

Phase 3 - Controlled pilot
8-16 WEEKS
- Approved site
- field dataset
- operator feedback
- VAPT and runbooks

Phase 4 - Production
AFTER ACCEPTANCE
- HA sector stack
- signed releases
- SLO operations
- C2 integration

Finale time	Primary output	Owner focus	Gate
0-4h	Contracts, Compose, streams and baseline demo	M1 + M5 + M6	One camera frame reaches UI
4-10h	Detector/tracker + model service	M2	FPS and prediction contract green
10-16h	ANPR/night candidate path	M3	Recorded day/night examples
16-22h	Fence/behavior/decision engine	M4	Persistence and false-alert tests
22-28h	Incidents, evidence, map and alerts	M5 + M6	Signed event visible end-to-end
28-32h	Offline queue, retry and recovery	M1 + M6	Link-loss replay passes
32-35h	Metrics, regression and judge scenarios	All	Acceptance dashboard frozen
35-36h	Demo reset, pitch rehearsal and backup	All	Two complete dry runs

Rule compliance

Prepare reusable components and tests only within the organizer's permitted scope. During the finale, keep an auditable commit history and demonstrate the work required by the official rules and mentors.

Judge Demo Story and Acceptance Gates

A five-minute product demonstration should prove value, measurement, resilience and honesty

1 Connect existing camera

Register an IP camera or replayed RTSP source; show codec, health, time sync and no dedicated smart hardware.

2 Detect and track

Show person and vehicle tracks with model version, confidence and live FPS/latency.

3 Trigger a virtual fence

A persistent track crosses a configured polygon and direction rule; single-frame noise is ignored.

4 Create incident evidence

Generate a priority incident with pre/post clip, keyframe, reason codes and SHA-256 manifest.

5 Break connectivity

Disable the command link; local alarm continues and event is queued without data loss.

6 Recover and audit

Restore the link; incident syncs once, appears on map, and audit trail shows every transition.

Gate	Pass condition	Evidence
Functional	All required SIH analytics represented in architecture; core vertical slice works	Live demo + recorded backup
Accuracy	No fabricated metric; frozen dataset and per-slice results available	Metrics report + confusion matrix
Latency	Critical event P95 measured under stated hardware/streams	Grafana or benchmark dashboard
Resilience	Local operation during link loss and replay after recovery	Automated integration test
Security	Authenticated device/operator, signed evidence and audit	Negative auth + hash verification
Scalability	Capacity assumptions tied to codec/FPS/model/hardware	Load test and sizing sheet

Closing line

Our innovation is not a single detector. It is a secure, offline-first and measurable intelligence layer that upgrades the CCTV infrastructure already deployed at remote border locations.

Target Repository Structure, Immediate Build Order and References

Use this structure as the contract between team members and AI coding agents

Recommended monorepo

```
VigilAI-Platform/
  apps/command-ui/               React + TypeScript operator HMI
  services/control-api/          cameras, zones, incidents, users, audit
  services/evidence-service/     clips, keyframes, hashes, retention
  edge/edge-agent/               ONVIF/RTSP, health, ring buffer, sync
  edge/vision-runtime/           detector, tracker, inference adapters
  edge/analytics/                anpr, face, night, behavior, fence
  packages/contracts/            Protobuf, JSON Schema, OpenAPI samples
  packages/security/             device identity, policy, auth helpers
  ml/datasets/                   manifests and DVC pointers, no secrets
  ml/training/                   reproducible train/evaluate/export jobs
  ml/model-registry/             model cards and promotion metadata
  deploy/compose/                SIH and single-post demo
  deploy/k3s/                    remote edge deployment
  deploy/kubernetes/             sector command deployment
  tests/contract/                schema and compatibility tests
  tests/integration/             camera-to-incident, link-loss, replay
  tests/performance/             FPS, latency, capacity and soak tests
  docs/adr/                      architecture decisions
  docs/runbooks/                 operations, recovery and security
```

Immediate build order

- 1. Freeze event schemas and golden sample payloads.
- 2. Create RTSP replay source and edge-agent health path.
- 3. Containerize current detector/tracker and publish TrackEvent.
- 4. Implement virtual-fence persistence and IncidentEvent.
- 5. Store evidence manifest and display incident in command UI.
- 6. Add link-loss queue, replay, metrics and acceptance report.

Current repository baseline

- GitHub: puriaditya004-max/VigilAI-Platform
- Audited commit: 0690c45 - traffic density and vehicle speed modules.
- Reuse patterns are valuable; production percentages in this blueprint are estimates.

Reference	Purpose
SIH26-26187 problem list	Exact problem scope, capabilities and expected solution
https://www.onvif.org/profiles/profile-t/	Advanced IP video streaming interoperability
https://docs.nvidia.com/metropolis/deepstream/dev-guide/	Production multi-stream video analytics runtime
https://docs.nvidia.com/deeplearning/triton-inference-server/user-guide/docs/	Versioned model serving, protocols, health and metrics
https://mlflow.org/docs/latest/ml/model-registry/	Model lineage, registry and lifecycle governance
https://opentelemetry.io/docs/	Vendor-neutral traces, metrics and logs

Final status: production target architecture - not a claim of completed field deployment, certification or measured 100% AI accuracy